Curriculum Vitae

Basic Information

Name in full : Minho Song
Office : Science hall #125, Yonsei University, Seoul, South Korea
Research Area : algebraic and enumerative combinatorics, knot theory

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Positions

March 2026 – Present	Post-doctoral fellow, Yonsei University.
September 2023 – February 2026	Research fellow, Sungkyunkwan University.
May 2019 – August 2023	Post-doctoral fellow, AORC, Sungkyunkwan University.
May 2018 – April 2019	BK Post-doctoral fellow, Sungkyunkwan University.
March 2018 – April 2018	Post-doctoral fellow, Sungkyunkwan University.

Education

March 2012 – February 2018 Ph.D. in Mathematics, Sungkyunkwan University.
Advisor: Prof. Gi-Sang Cheon
Thesis : Riordan Lie Algebra and Extended Riordan groups

March 2010 – February 2012 M.S. in Mathematics, Sungkyunkwan University.
Advisor: Prof. Gi-Sang Cheon
Thesis : Geometric Progression Matrices and Generalizations

March 2006 – February 2010 B.S. in Mathematics, Sungkyunkwan University.

Research Interests

My research is in algebraic and enumerative combinatorics, broadly construed. My current interests include Coxeter and Catalan combinatorics, Hecke and skein algebras, and their connections with knot theory. I also work on symmetric functions and representation theory, orthogonal polynomials, Riordan groups, and algebraic graph theory.

A central theme of my work is to understand how combinatorial models reflect algebraic, geometric, and topological phenomena. I am particularly interested in connections among combinatorial structures, representation-theoretic frameworks, and knot invariants.

I also explore probabilistic aspects and applications of combinatorics. In this direction, I have studied questions related to Kemeny's constant for finite Markov chains.

More recently, I have developed an interest in artificial intelligence for mathematics. In particular, I am interested in developing and applying FunSearch-style approaches based on AlphaEvolve as computational tools for conjecture discovery and structural exploration in mathematics.

Research Activities

Preprints

- *An explicit formula for Koornwinder moments and Rains' positivity conjecture* (with Younggwang Cho, Donghyun Kim, Jang Soo Kim, Hojoon Lee, and Jing Liu), arXiv:2606.14241.
- *Combinatorics of generalized orthogonal polynomials of type R_{II}* (with Jang Soo Kim), arXiv:2411.12345.

Published/Accepted Papers

- *Combinatorics of orthogonal polynomials on the unit circle* (with Jihyeug Jang), The Ramanujan Journal, 68:84 (2025), 1–36.
- *Combinatorial reciprocity for Riordan arrays* (with Jihyeug Jang and Louis W. Shapiro), Linear Algebra and its Applications, 721 (2025), 419–448.
- *Refined canonical stable Grothendieck polynomials and their duals, Part 2* (with Byung-Hak Hwang, Jihyeug Jang, Jang Soo Kim, and U-Keun Song), European Journal of Combinatorics, 127 (2025), 104166.
- *Kemeny's constant and enumerating Braess edges in trees* (with Jihyeug Jang, Mark Kempton, Sooyeong Kim, Adam Knudson, and Neal Madras), Linear & Multilinear Algebra, 73(8) (2025), 1529–1565.
- *Strictly monotone sequences of lower and upper bounds on Perron values and their combinatorial applications* (with Sooyeong Kim), Linear & Multilinear Algebra, 73(2) (2025), 322–350.
- *Refined canonical stable Grothendieck polynomials and their duals, Part 1* (with Byung-Hak Hwang, Jihyeug Jang, Jang Soo Kim, and U-Keun Song), Advances in Mathematics, 446 (2024), 109670.
- *A Riordan group poset* (with Louis W. Shapiro), Linear Algebra and its Applications, 689 (2024), 66–85.
- *Kemeny's constant and Wiener index on trees* (with Jihyeug Jang and Sooyeong Kim), Linear Algebra and its Applications, 674 (2023), 230–243.
- *Negative moments of orthogonal polynomials* (with Jihyeug Jang, Donghyun Kim, Jang Soo Kim, and U-Keun Song), Forum of Mathematics, Sigma 11 (2023), Paper No. e22, 34 pp.
- *Enumeration of bipartite non-crossing geometric graphs* (with Gi-Sang Cheon, Hong Joon Choi, Guillermo Esteban), Discrete Applied Mathematics, 317(2022), 86–100.
- *A new aspect of Riordan arrays via Krylov matrices* (with Gi-Sang Cheon), Linear Algebra and its Applications, 554 (2018), 329–341.
- *Finite and infinite dimensional Lie group structures on Riordan groups* (with Gi-Sang Cheon, Ana Luzon, Manuel A. Moron, Luis Felipe Prieto-Martinez), Advances in Mathematics, 319 (2017), 522–566.

Workshops Attended

- *Thematic Program on AI and Mathematics*, Korea Institute for Advanced Study, August 3–7, 2026.
- *PACOH: Artificial Intelligence for Combinatorics*, Jeju, July 6–10, 2026.
- *2025 KIAS Winter School on Mathematics and AI*, Jeongseon, December 2–5, 2025.
- *Mathematical Modeling in Industry XIX*, University of Minnesota, August 5–14, 2015.

Conference Talks

- *Deograms and their links to rational Catalan objects*, ASIACOMB 2026, August 24–28, 2026, Daejeon, South Korea.
- *Moments and lattice paths for orthogonal polynomials of type R_{II}*, The 18th International Symposium on Orthogonal Polynomials, Special Functions and Applications (OPSFA-18), August 17–21, 2026, Kyoto, Japan.
- *Exact enumeration via Riordan arrays*, Soongsil University Mathematics Colloquium, April 29, 2026, Soongsil University, South Korea.
- *An invitation to Riordan group theory*, Discrete Analysis Seminar, April 7, 2026, Yonsei University, South Korea.
- *A family of pseudo-involutory functions*, Joint Mathematics Meetings 2026, January 3–7, 2026, Washington, USA.
- *A unified combinatorial framework for orthogonal polynomials including type R_{II}*, The 34th KIAS Combinatorics Workshop, May 29–31, 2025, Jeju, South Korea.
- *Two Rational Catalan Objects*, 2025 KSIAM Spring Conference, May 16–18, Konkuk University, South Korea.
- *Combinatorics of generalized orthogonal polynomials of type R_{II}*, 2025 KMS Spring Meeting, April 24–26, 2025, KAIST, South Korea.
- *Combinatorics of the orthogonal polynomials on the unit circle (OPUC)*, 2024 Combinatorics Workshop, August 28–30, 2024, Chungbuk National University, South Korea.
- *About Riordan group posets*, The 9th International Symposium on Riordan Arrays and Related Topics (9RART), June 3–5, 2024, Howard University, USA.
- *Combinatorial reciprocity for Riordan arrays*, The 30th KIAS Combinatorics Workshop, March 8–9, 2024, KIAS, South Korea.
- *Combinatorics on the negative part of Riordan matrices*, The 25th Conference of the International Linear Algebra Society, June 12–16, 2023, Madrid, Spain.
- *Enumerative results for connected bipartite non-crossing geometric graphs*, The 24th Conference of the International Linear Algebra Society, June 20–24, 2022, Ireland.

- *Enumeration of connected bipartite non-crossing geometric graphs*, 2021 KMS Spring Meeting, April 29–30, 2021, South Korea (Online).
- *Riordan group theory and connected bipartite non-crossing graphs*, 2020 Gyeonggi-Incheon Combinatorics Workshop, November 27, 2020, South Korea (Online).
- *A relation between strongly connected tournaments and connected graphs*, The 6th International Workshop on Riordan Arrays and Related Topics, July 1–5, 2019, TSIMF, Sanya, China.
- *A proof of an open problem on the commutator subgroup of the Riordan group*, Joint Mathematics Meetings, January 16–19, 2019, Baltimore, USA.
- *A new aspect of the Riordan group via Krylov matrices*, The 3rd International Symposium on Riordan Arrays and Related Topics, June 20–23, 2016, Illinois Wesleyan University, USA.
- *Lie Theory in the Riordan group*, East Asian Core Doctoral Forum on Mathematics, January 18–21, 2015, National Taiwan University, Taiwan.
- *Krylov matrices versus orthogonal polynomials*, The Second International Symposium on Riordan Arrays and Related Topics, July 14–16, 2015, Campus Lecco, Politecnico di Milano, Italy.
- *The Riordan Lie algebra*, 2014 KMS Annual Meeting, October 24–25, 2014, Yonsei University, South Korea.
- *On the commutator subgroup of the Riordan group*, The 19th Conference International Linear Algebra Society, August 6–9, 2014, Sungkyunkwan University, South Korea.
- *The Riordan Lie group and the corresponding Lie algebra*, 2014 Kangwon-Kyungki Mathematical Society Meeting, June 20, 2014, Incheon National University, South Korea.
- *Geometric progression matrices and an application to the orthogonal polynomials*, 2014 AKOOS-PNU International Conference, February 5–7, 2014, Pusan National University, South Korea.

Academic Event Organization

- Co-Organizer, *Intensive Workshop on Algebraic Combinatorics*, Department of Mathematics, Sungkyunkwan University, January 14–18, 2025.
- Co-Organizer, *Intensive Workshop on Enumerative Combinatorics*, Department of Mathematics, Sungkyunkwan University, January 24–28, 2022.

Honors and Awards

- *Sejong Science Fellowship* from National Research Foundation of Korea, March 2022 – February 2027.
- *Creativity Challenge Research* from National Research Foundation of Korea, June 2019 – May 2022.
- *Doctoral Dissertation Award* from Korean Mathematical Society, April 21, 2018.
- *2014 AKOOS-PNU International Conference Best Presentation Award* from Pusan National University, February 7, 2014.
- *BK21 Scholarship Program for graduate students* from National Research Foundation of Korea, March 2010 – February 2016.
- *Global Ph.D Fellowship* from National Research Foundation of Korea, March 2010 – February 2013.

Date: August 13, 2026